Formula 1 Big Data Analytics Pipeline
M4 Final Report

Saja Plath

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1 Executive Summary

This project implements a scalable Big Data pipeline for Formula 1 race analytics using Apache Spark and the FastF1 API. The system ingests multi-season telemetry and race data, processes it using distributed transformations, and generates structured analytical outputs.

The goal is to demonstrate an end-to-end data engineering workflow that includes ingestion, transformation, validation, and output generation. The pipeline is designed to be reproducible, scalable, and representative of real-world data engineering systems.

2 Problem Statement

Formula 1 generates large volumes of telemetry and race data that are difficult to analyze manually. This project addresses the need for a scalable pipeline that can process multi-season racing data and extract meaningful performance insights such as driver consistency, lap performance, and race comparisons.

3 System Architecture

The pipeline follows a multi-stage architecture:

- Data Ingestion (FastF1 + OpenF1 APIs)
- Data Processing (Apache Spark)
- Feature Engineering (Lap time and performance metrics)
- Validation Layer (Data quality checks)
- Output Layer (Parquet datasets)

3.1 Pipeline Flow

FastF1 API → Ingestion → Spark Processing → Feature Engineering → Validation → Output

4 Implementation Details

4.1 Data Ingestion

The ingestion layer collects:

- Session metadata from OpenF1
- Lap timing data from FastF1
- Race results and telemetry

Data is collected across multiple seasons to ensure scalability and realistic workload simulation.

4.2 Data Processing

Apache Spark is used for distributed processing. Key transformations include:

- Dropping null values
- Converting lap times into numerical formats
- Grouping by driver and race
- Computing aggregate statistics (mean, min, max)

4.3 Feature Engineering

New derived metrics include:

- Average lap time per driver
- Fastest lap per race
- Race-level pace comparisons

5 Validation and Data Quality

Data validation ensures correctness and reliability of outputs.

5.1 Validation Checks

- Null value detection on critical fields
- Empty dataset checks
- Schema validation
- Statistical sanity checks on lap times and speeds

5.2 Known Limitations

- Data availability depends on FastF1 API
- Local Spark performance varies by machine
- Cached data is excluded from version control

6 Results

The pipeline successfully processes multi-season Formula 1 data and generates structured outputs.

6.1 Sample Output Summary

- OpenF1 processed records: ∼120
- FastF1 lap records: ∼90
- FastF1 results records: ∼22

6.2 Example Insights

- Driver performance varies significantly across sessions
- Top drivers maintain consistent lap times across races
- Race pace is strongly influenced by team strategy and conditions

7 Performance Considerations

The pipeline is designed for scalability using Spark. Performance depends on:

- Dataset size (multi-season expansion improves realism)
- Local compute resources
- API response latency

8 Conclusion

This project demonstrates a complete Big Data pipeline for Formula 1 analytics. It successfully integrates data ingestion, distributed processing, and validation into a unified system.

The final system is modular, scalable, and suitable for extension into real-time analytics or cloud deployment.

9 Future Work

- Expand to full historical F1 dataset
- Add real-time streaming ingestion
- Deploy on cloud platforms (AWS / Databricks)
- Integrate machine learning for performance prediction
- Add visualization dashboard layer